Home Work #1

N1. $32, 16, 8, 4, 2, 1$
$\quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1$

- $0_{10} = 000000_2$
- $13_{10} = 8_{10} + 4_{10} + 1_{10} = 0 \cdot 32 + 0 \cdot 16 + \underline{1} \cdot 8 + \underline{1} \cdot 4 + 0 \cdot 2 + \underline{1} \cdot 1 = 001101_2$
- $24_{10} = 16_{10} + 8_{10} = 0 \cdot 32 + 1 \cdot 16 + 1 \cdot 8 + 0 \cdot 4 + 0 \cdot 2 + 0 \cdot 1 = 011000_2$
- $63_{10} = 32 + 16 + 8 + 4 + 2 + 1 = 111111_2$ → max to attain for unsigned

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- $16_{10} = 16 = 0 \cdot 32 + 1 \cdot 16 + 0 \cdot 8 + 0 \cdot 4 + 0 \cdot 2 + 0 \cdot 1 = 010000_2$
- -2_{10} $\quad 2_{10} = 1 \cdot 2 = 000010_2$

  inverse $= 111101_2$

  inv + 1 : 111110_2

  $-2_{10} = 111110_2$

- $31_{10} = 16 + 8 + 4 + 2 + 1 = 011111_2$
- -32_{10} → the minimal to attain, by rule it is 100000_2

  $\parallel$

  100000_2

N2.

unsigned : $\sum_{i=0}^{5} b_i \cdot 2^i$

two's complement : $-b_5 \cdot 2^5 + \sum_{i=0}^{4} b_i \cdot 2^i$

- 000101

- unsigned: $0 \cdot 32 + 0 \cdot 16 + 0 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 5_{10}$
- tc: $-0 \cdot 32 + 0 \cdot 16 + 0 \cdot 8 + 1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 5_{10}$

- 101011

- unsigned: $1 \cdot 32 + 1 \cdot 8 + 1 \cdot 2 + 1 \cdot 1 = 43_{10}$
- tc: $-1 \cdot 32 + 1 \cdot 8 + 1 \cdot 2 + 1 \cdot 1 = -21_{10}$

- 111111
- unsigned: 63_{10} (max possible, check N1)
- tc: $-32+16+8+4+2+1 = -1_{10}$

- 100000
- unsigned: 32_{10} rest is 0
- signed: -32_{10}

N3.

- $7 \,|\, 16$, 0 $\Rightarrow$ $7 = 07_{16}$

  7 - remainder

- $240 \,|\, 16$, -15, $15 \,|\, 16$, 0 $\Rightarrow$ $240 = F0_{16}$

  15 - remainder
  ||
  F

- $171 \,|\, 16$, -160, $10 \,|\, 16$, 0 $\Rightarrow$ $171 = AB_{16}$

  11 remainder

  10 - remainder

  10, 11
  ↓ ↓
  A B

- $126 \,|\, 16$, -112, $7 \,|\, 16$, 0 $126 = 7E_{16}$

  14 remainder

  7 remainder

  7, 14
  ↓
  E

N4.

- 0x3C

  3 - 0011 $\quad$ 0x3C = $0011\ 1100_2$

  C - 1100

- 0x7E

  7 - 0111 $\quad$ 0x7E = $0111\ 1110_2$

  E - 1110

- 0xFF = $1111\ 1111_2$

  F - 1111

- 0xA5 = $1010\ 0101_2$

  A - 1010

  5 - 0101_2

  $1 \cdot 4 \quad 1 \cdot 1 = 4+1=5$

N5.

- 0011 1100 = 0xC4

  inv: 1100 0011

  inv+1: $\underbrace{1100}_{C}\underbrace{0100}_{4}$

- 0111 1110 = 0x82

  inv: 1000 0001

  inv+1: $\underbrace{1000}_{8}\ \underbrace{0010}_{2}$

- 1111 1111 = 0x01

  inv: 00000000

  inv+1: $\underbrace{0000}_{0}\ \underbrace{0001}_{1}$

- 1010 0101 = 0x5B

  inv: 01011010

  inv+1: $\overbrace{0101}^{5}\ \overbrace{1011}^{B}$

N6. 0xDEADBEEF

DE, AD, BE, EF

Big-endian – most significant byte at the lowest memory address

order: DE, AD, BE, EF

Little-endian – least significant byte at the lowest memory address

order: EF, BE, AD, DE

N7. 5bit:

$7_{10} \rightarrow 00111 = 1\cdot1+2\cdot1+4\cdot1 = 7$

$15_{10} \rightarrow 01111 = 1\cdot1+2\cdot1+4\cdot1+8\cdot1 = 7+8 = 15$

$-16_{10} \rightarrow 10000 = -1\cdot16+0$

$-5_{10} \rightarrow 11011 = -1\cdot16+1\cdot8+1\cdot2+1\cdot1 = -16+11 = -5$

sign-extend:

7 → 00000111

15 → 00001111

-16 → 11110000

-5 → 11111011

zero-extend:

7 → 00000111

15 → 00001111

-16 → 00010000

-5 → 00011011

N8.

Unsigned

- 7 → 0111 = 4+2+1
  9 → 1001 = 8+1

$$\begin{array}{r} 0111 \\ +\ 1001 \\ \hline 10000 \end{array}$$

4 bit, take mod => Answer: 0000

- Signed

4 → 0100

-5 5 → 0101

inv: 1010

inv+1: 1011 = -5

$$\begin{array}{r} 0100 \\ +\ 1011 \\ \hline 1111 \end{array}$$

Answer: 1111